

Data Cleaning, Transformation and Modelling

Question 1: What is Data Loading in Power BI and why is it considered the first step of analysis?

Answer:

Data Loading is the process of importing data from different sources (Excel, CSV, SQL Server, Web, etc.) into Power BI. It is the first step because data must be available in Power BI before it can be cleaned, transformed, modeled, or visualized.

Question 2: Explain the difference between "Load" and "Transform Data" in Power BI.

Answer:

Load	Transform Data
Imports data directly into Power BI.	Opens Power Query Editor to clean and modify data before loading.
No changes are made to the data.	Allows filtering, removing duplicates, replacing null values, changing data types, etc.
Faster process.	Used for data preparation.

Question 3: What is a Fact Table and a Dimension Table? Give examples from the dataset.

Answer:

Fact Table:

- Contains measurable values or business data.
- Example: Corona_Virus_2020 (Confirmed, Deaths, Recovered, Active Cases)

Dimension Table:

- Contains descriptive information.
 - Example: Vaccination_Status, Country, State, Date
-

Question 4: Why is Star Schema preferred over Snowflake Schema in Power BI?

Answer:

Data Cleaning, Transformation and Modelling

- Faster query performance.
 - Simpler relationships.
 - Better report performance.
 - Easy to understand and maintain.
 - Reduces complex joins.
-

Question 5: Identify and remove duplicate records based on Date, Country, and State.

Answer:

1. Open **Power Query Editor**.
 2. Select **Date**, **Country**, and **State** columns.
 3. Click **Home** → **Remove Rows** → **Remove Duplicates**.
 4. Click **Close & Apply**.
-

Question 6: Identify and replace null values in Vaccination_Status.

Answer:

1. Open **Power Query Editor**.
 2. Select the **Vaccination_Status** column.
 3. Choose **Replace Values**.
 4. Replace **null** with **"Not Vaccinated"** (or another appropriate value).
 5. Click **Close & Apply**.
-

Question 7: Create a new column to calculate Recovery Rate.

Answer (DAX):

Recovery Rate =
DIVIDE(
 Corona_Virus_2020[Recovered],
 Corona_Virus_2020[Confirmed],
 0
) * 100

Question 8: Create a summarized table showing total confirmed cases by Country.

Data Cleaning, Transformation and Modelling

Answer (DAX):

Country Summary =

SUMMARIZE(

Corona_Virus_2020,

Corona_Virus_2020[Country],

"Total Confirmed",

SUM(Corona_Virus_2020[Confirmed])

)